The documentation and process conversion measures necessary to comply with this revision shall be completed by 6 February 2007.

INCH POUND

MIL-PRF-19500/228K
6 November 2006
SUPERSEDING
MIL-PRF-19500/228J
16 July 2001

* PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, RECTIFIER, TYPES 1N3611, 1N3612, 1N3613, 1N3614, 1N3957, 1N3611EG1, 1N3612EG1, 1N3613EG1, 1N3614EG1, 1N3957EG1, JAN AND JANTX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for 1.0 ampere silicon rectifier diodes. Two levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1, similar to DO-41, and figure 2, epoxy-glass case outline.

* 1.3 Maximum ratings. Unless otherwise specified T_A = 25°C.

Types	V_{RWM}	I_O		I_{FSM}	Barometric pressure (reduced)	T_{STG} and T_J	$R_{\theta JL}$ L = .375 inch (9.53 mm)
		At: T_A = +100°C (1) (2)	At: T_A = +150°C (1) (2)	T_A = +25°C I_O = 1 A dc t_p = 8.0 ms			
	V(pk)	A dc	mA dc	A(pk)	mmHg	°C	°C/W
1N3611, EG1 (3)	200	1	300	30	8	-65 to +175	38
1N3612, EG1 (3)	400	1	300	30	8	-65 to +175	38
1N3613, EG1 (3)	600	1	300	30	8	-65 to +175	38
1N3614 , EG1 (3)	800	1	300	30	54	-65 to +175	38
1N3957, EG1 (3)	1,000	1	300	30	87	-65 to +175	38

(1) From I_O rating is independent of heat sinking, special mounting, or leads of the device.
(2) Derate linearly at 13.3 mA between T_A = +100°C and T_A = +175°C.
(3) See 3.7 herein.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

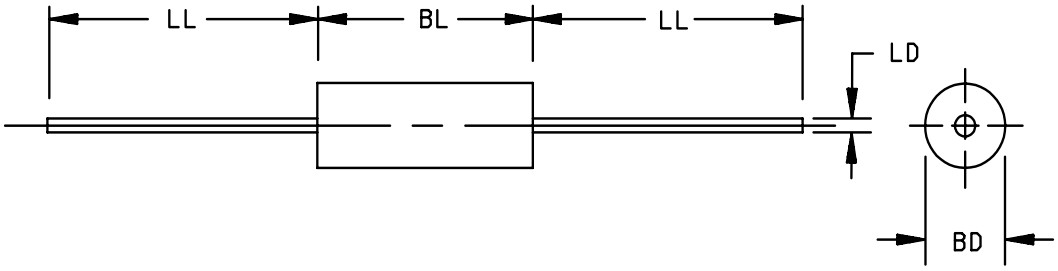
DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil. or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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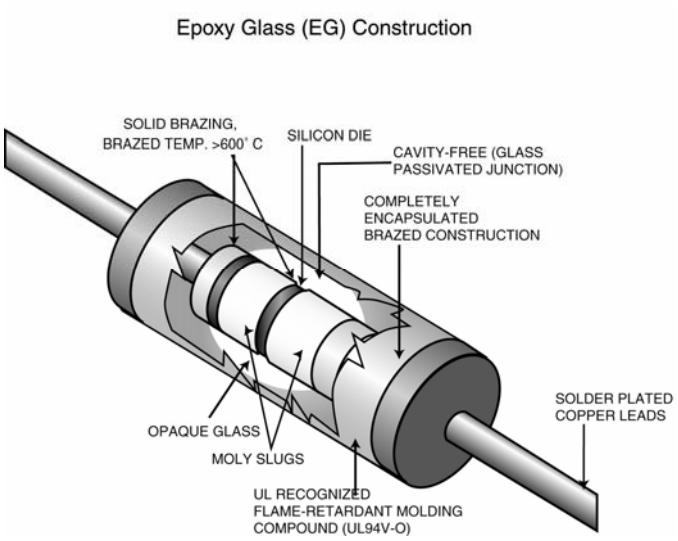


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BD	.060	.110	1.52	2.79	
BL	.140	.205	3.56	5.21	
LD	.025	.034	0.64	0.86	3
LL	.600	1.500	15.24	38.10	4

NOTES:
1. Dimensions are in inches.
2. Millimeter equivalents are given for general information only.
3. Dimension LD shall be measured at the largest diameter.
4. Dimension BD shall include the sections of the lead over which the diameter is uncontrolled.
   This uncontrolled area is defined as the zone between the edge of the diode body and extending
   .050 inch (1.27 mm) onto the leads.
5. See note in 3.7 herein.

* FIGURE 1. Physical dimensions.

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NOTE: See 3.7 herein.

FIGURE 2. Epoxy-Glass (EG1 case outline) construction (for reference only).

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3. REQUIREMENTS

* 3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

EG - Epoxy over glass hermetically sealed EG1 case.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 herein. No lead (Pb) shall be used in the construction of the die bonds.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Diode construction. These devices shall be constructed utilizing non-cavity double plug construction with high temperature bonding between both sides of the silicon die and terminal pins. Figure 2, epoxy-glass (EG1 case outline) construction is for reference only.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, and table I.

* 3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I.

3.7 Marking. Devices shall be marked as specified in MIL-PRF-19500.

NOTE: The EG1 version, epoxy over glass construction, in this specification is a construction identifier. It denotes a type of design and not a change in package outline. EG1 is not intended to be a package outline identifier. The EG1 version meets all of the requirements of a non-EG1 version including case outline, hermeticity, and electricals parameters. The EG1 may be marked and shipped as a non-EG1 part number because it is an equivalent device in terms of form, fit, and function. The EG1 is interchangeable and substitutable for the non-EG1 version.

3.8 Polarity. The polarity shall be indicated with a contrasting color band to denote the cathode end or the use of other techniques considered commercial practice.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

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4.2.1 Group E qualification. Group E qualification shall be performed herein for qualification or re-qualification only. In case qualification was awarded to a prior revision of the associated specification that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot to this revision to maintain qualification.

* 4.3 Screening (JANTX level only). Screening shall be in accordance with appendix E, table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table E-IV of MIL-PRF-19500)	Measurement JANTX level
(1) 3c	See 4.3.2
9	Not applicable
11	I_{R1} and V_{F1}
12	See 4.3.1
(2) 13	Subgroup 2 of table I herein. ΔI_{R1} = 100 percent of initial reading or ± 50 nA dc, whichever is greater. ΔV_{F1} = ± 0.1 V dc.

(1) Shall be performed anytime after temperature cycling, screen 3a, and does not need to be repeated in screening requirements.
(2) For clear glass diodes, the hermetic seal (gross leak) may be performed at anytime after temperature cycling.

* 4.3.1 Power burn-in conditions. Power burn-in conditions are as follows (see 4.5.1): $I_{O(min)}$ = rated I_O (see 1.3 herein). T_A = 55°C maximum. Test conditions in accordance with method 1038 of MIL-STD-750, condition B. Adjust I_O or T_A to achieve the required T_J. T_J = 135°C minimum, 175°C maximum. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, mounting conditions) may be used for JANTX quality level. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

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* 4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 as applicable of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW} (V_C and V_H where appropriate). Measurement delay time (t_{MD}) = 70 μs max. The thermal impedance limit used in screen 3c and table I, subgroup 2 shall be set statistically by the supplier. See table II, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table E-V of MIL-PRF-19500 and table I herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VIb (JANTX, and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with the applicable steps of table I, subgroup 2 herein. Delta requirements shall be in accordance with the applicable steps of table III herein.

* 4.4.2.1 Group B inspection, appendix E, table E-VIb of MIL-PRF-19500.

Subgroup	Method	Conditions
B2	4066	I_{FSM} = 30 A(pk), 10 surges of 8.3 ms each at 1 minute intervals, I_O = 1 A dc, V_{RWM} = rated V_{RWM} (see 1.3), T_A = +25°C. Sampling plan = 10.
B3	1026	T_A = +55°C (max), I_O = 300 mA, V_R = rated V_{RWM} (see 1.3), t = 340 hours, f = 60 Hz. Adjust I_O to achieve a T_J = +150°C minimum.
B5	4081	+25°C ≤ T_L ≤ +35°C (recorded before test is performed), $R_{\theta JL}$ ≤ maximum rated $R_{\theta JL}$ (see 1.3 and 4.3.2).
B6	1032	T_A = +175°C, t = 340 hours.

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the applicable steps of table I, subgroup 2 herein. Delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.3.1 Group C inspection appendix E, table E-VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	2036	Test condition A, weight = 4 pounds, t = 15 s.
C2	2036	Test condition E.
C5	1001	V_{RWM} = rated V_{RWM} (see 1.3), pressure = rated pressure (see 1.3), t = 1 minute, sampling plan = 15.
C6	1026	T_A = +55°C (max), I_O = 300 mA, V_{RWM} = rated V_{RWM} (see 1.3), f = 60 Hz. Adjust I_O to achieve a T_J = +150°C minimum.

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* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and table II herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

* 4.5.1 Free air power burn-in and life tests. The use of a current limiting or ballast resistor is permitted provided that each DUT still sees the full I_O (minimum) and that the minimum applied voltage, where applicable, is maintained through out the burn-in period. Use method 3100 of MIL-STD-750 to measure T_J.

4.5.2 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance	3101	See 4.3.2	$Z_{\theta JX}$			°C/W
Forward voltage	4011	I_F = 1 A dc (pulsed) (see 4.5.2); duty cycle = 2 percent maximum.	V_{F1}	.6	1.1	V pk
Reverse current leakage	4016	V_R = rated V_{RWM} (see 1.3), dc method.	I_{R1}		1.0	µA dc
Subgroup 3						
High temperature operation:		T_A = 150°C.				
Reverse current leakage	4016	V_R = rated V_{RWM} (see 1.3), dc method.	I_{R2}		300	µa dc
Low temperature operation:		T_A = -65°C.				
Forward voltage	4011	I_F = 1.0 A dc (pulsed) (see 4.5.2).	V_{F2}	0.6	1.5	V dc

1/ For sampling plan, see MIL-PRF-19500.

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* TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			22 devices c = 0
Thermal shock	1056	100 cycles, 0° to 100°C.	
Hermetic seal	1071	Test condition C and H1	
Electrical measurement		See table I, group A, subgroup 2.	
Subgroup 2			22 devices c = 0
Steady-state reverse bias	1038	Test condition A, 1,000 hours.	
Electrical measurement		See table I, subgroup 2.	
Subgroup 4			Sample size = N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			
Not applicable			

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* TABLE III. Groups A, B, and C electrical measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	I_F = 1.0 A dc pulse width = 8.3 ms maximum, duty cycle = 2 percent maximum.	V_{F1}	0.6	1.1	V dc
2.	Reverse current	4016	DC method; V_R = rated (see 1.3).	I_{R1}		1.0	µA dc
3.	Forward voltage	4011	I_F = 1.0 A dc pulse width = 8.3 ms maximum; duty cycle = 2 percent maximum.	ΔV_{F2}	±0.1 V dc maximum		
4.	Reverse current	4016	DC method; V_R = rated (see 1.3).	ΔI_{R1}	±50 nA dc or 100 percent of initial value, whichever is greater.		

1/ The electrical measurements for appendix E, table E-VIb (JAN and JANTX) of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table III herein, steps 1 and 2.
   b. Subgroup 3, see table III herein, steps 1, 2, 3, and 4.
   c. Subgroup 6, see table III herein, steps 1 and 2.
   d. Subgroup 7, see table III herein, steps 1 and 2.

2/ The electrical measurements for appendix E, table E-VII of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table III herein, steps 1 and 2.
   b. Subgroup 3, see table III herein, steps 1 and 2.
   c. Subgroup 5, see table III herein, step 2.
   d. Subgroup 6, see table III herein, steps 1, 2, 3, and 4.

3/ The electrical measurements for table E-IX of MIL-PRF-19500 are as follows:
   a. Subgroup 1, see table III herein, steps 1, 2, 3, 4, and 5.
   b. Subgroup 2, see table III herein, steps 2, 3, 4, and 5.

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* 5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

* 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

* 6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.5.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, AV, MI, SM
Navy - AS, MC
Air Force - 19, 70, 99

Preparing activity:
DLA - CC

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